CAGE-Fake: Article Trustworthiness Audit Report

Calibration-Aware Gating and Evidence-Grounded Reasoning for Trustworthy Fake News Triage

LEGAL & METHODOLOGICAL DISCLAIMER: *A trained fake-news classifier predicts linguistic, stylometric, and semantic patterns learned from benchmark datasets; it does **NOT** independently verify whether a news story is factually true. URL extraction, model pattern recognition, and external fact verification remain strictly separate processes. High uncertainty cases are flagged for professional human editorial review.*

Source / URL: Manual Text Submission
Title: Senate Passes Historic Clean Energy Infrastructure Deal
Word Count: 42 words | **Date:** 2026-10-19 10:54:28

VERDICT: UNCERTAIN
Calibrated Confidence: 86.4%
Model Disagreement (std): 0.237
Gating Trigger: High Disagreement

1. Multi-Model Disagreement Analysis (5 Models)

Model Name	P(FAKE)	Model Vote	Confidence
Random Forest	52.6%	FAKE	52.6%
XGBoost	48.7%	REAL	51.3%
LightGBM	60.6%	FAKE	60.6%
Logistic Regression	0.0%	REAL	100.0%
CAGE-Fake Stack	13.6%	REAL	86.4%

Consensus Status: High Dissent / Conflicting Predictions
Inter-Model Standard Deviation: 0.2379 (Threshold Delta = 0.20)
Vote Distribution: 2 FAKE vs 3 REAL
Diagnostic: Inter-model disagreement (0.238) exceeds threshold (0.200).

2. News Credibility Fingerprint (Linguistic & Stylistic Scorecard)

Credibility Dimension	Score (0-100)	Journalistic Assessment
Sensationalism Discipline	100.0/100	Excellent
Source Attribution	0.0/100	Deceptive / Alert
Epistemic Grounding	100.0/100	Excellent
Punctuation Discipline	100.0/100	Excellent
Readability Structure	100.0/100	Excellent
Lexical Diversity	70.0/100	Adequate

Overall Credibility Index: 72.0/100
Credibility Tier: B (Standard News)

Scores are normalized against verified newsroom style guides: high attribution, balanced syntax, and low sensationalism maximize the index.

3. Transparent Reasoning & Counterfactual Sensitivity

Diagnostic Explanation:
1. Decision & Triage Status: The CAGE-Fake pipeline reached a verdict of **UNCERTAIN** with **86.4% confidence**.
- **Gating Trigger (Inter-Model Conflict)**: The ensemble exhibited significant dissent across the 5 models (std=0.238). Because models disagree, the system abstains from a definitive label to protect against false positives.

Counterfactual Simulation: The article already employs restrained factual syntax. Adding speculative adjectives or unanchored claims would increase deceptive risk indicators towards FAKE.
Sensitivity Delta: P(FAKE) changed from 13.6% to 13.6% (+0.0%) when neutralizing sensational tokens.

4. Local Linguistic Signal Attributions

Top FAKE-Driving Indicators	Weight / Impact	Top REAL-Driving Anchors	Weight / Impact
p_xgb	+6.964	p_rf	-7.063
disagreement	+2.528	feature_5	-4.794

feature_10	+2.477	p_lgbm	-3.914
feature_11	+2.229	feature_8	-3.320

5. Empirical Benchmark Model Comparison (Held-out Test Split)

Model	Accuracy	Macro-F1	ECE	Coverage	Selective Acc
Random Forest	0.8639	0.8635	0.1548	100.0%	0.8639
XGBoost	0.8770	0.8769	0.0380	100.0%	0.8770
LightGBM	0.8689	0.8688	0.0418	100.0%	0.8689
Logistic Regression	0.8967	0.8967	0.0381	100.0%	0.8967
Uncalibrated Stack	0.8984	0.8983	0.0330	100.0%	0.8984
CAGE-Fake (Calibrated + Gated)	**0.9000**	**0.8999**	**0.0362**	**84.4%**	**0.9476**

6. Research Calibration & Disagreement Visualizations







